Deep Dive in FlutterNEXT

GDG DevFest
Pisa 2023

Chi sono

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FlutterNEXT in 4 punti

Performance

Integration

New Architectures

Focus Dev
Experience

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Flutter

Dart 3

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Dart 3

Record

Record

```
class LatLng {
  final double lat;
  final double long;

  LatLng(this.lat, this.long);
}

LatLng geoLocation(String name) {
  if (name == 'Nairobi') {
    return LatLng(-1.2921, 36.8219);
  } else {
    ...
  }
}


void main(List<String> arguments) {
  final coords= geoLocation('Nairobi');
  print('Current location: ${cords.lat}, ${cords.long}');
}
```

Record

```dart
(double x, double y) geoLocation(String name) {
  if (name == 'Nairobi') {
    return (-1.2921, 36.8219);
  } else {
    ...
  }
}


void main(List<String> arguments) {
  final (lat, long) = geoLocation('Nairobi');
  print('Current location: $lat, $long');
}
```

Pattern

Pattern

```
sealed class Shape {}

class Square implements Shape {
  final double length;
  Square(this.length);
}

class Circle implements Shape {
  final double radius;
  Circle(this.radius);
}

double calculateArea(Shape shape) {
  if(shape is Square){
    return shape.l*shape.l;
  }else if(shape is Circle){
    math.pi * shape.r * shape.r
  }else{
    return 0;
  }
}
```

Pattern

```
sealed class Shape {}

class Square implements Shape {
  final double length;
  Square(this.length);

}

class Circle implements Shape {
  final double radius;
  Circle(this.radius);
}

double calculateArea(Shape shape) => switch (shape) {
  Square(length: var l) => l * l,
  Circle(radius: var r) => math.pi * r * r
};
```

Direct platform library interop

Direct platform library interop

jnigen

Java

Kotlin

ffigen

Objective-C

Swift

Direct platform library interop

jnigen

Java

Kotlin

```java
package dev.dart;

public class Example {
  public static int sum(int a, int b) {
    return a + b;
  }
}
```

Direct platform library interop

jnigen

Java

Kotlin

```yaml
jnigen:
output:
  c:
    library_name: example
    path: src/example/
  dart:
    path: lib/example.dart
    structure: single_file

source_path:
  - 'java/'
classes:
  - 'dev.dart.Example'
```

Direct platform library interop

jnigen

Java

Kotlin

```dart
import 'package:jni/jni.dart';
import 'package:jnigen_example/example.dart';
import 'package:path/path.dart';

void main(List<String> args) {
  Jni.spawn(
    dylibDir: join('build', 'jni_libs'),
    classPath: ['java'],
  );

  try {
    final a = int.parse(args[0]);
    final b = int.parse(args[1]);
    print(Example.sum(a, b)); // prints a + b
  } on FormatException catch (_) {
    print('The arguments must be integers.');
    exit(1);
  }
}
```

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Direct platform library interop

jnigen

Java

Kotlin

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Objective-C

Swift

Direct platform library interop

ffigen

Objective-C

Swift

```
import Foundation

@objc public class Example : NSObject {
  @objc public func sum(Int a, Int b) -> Int {
    return a + b;
  }

  @objc public var someField = 123;
}
```

Direct platform library interop

ffigen

Objective-C

Swift

```yaml
ffigen:
  name: SwiftLibrary
  description: Bindings for swift_api.
  language: objc
  output: 'swift_api_bindings.dart'
  exclude-all-by-default: true
  objc-interfaces:
    include:
      - Example'
    module:
      'Example': 'swift_module'
  headers:
    entry-points:
      - 'swift_api.h'
  preamble: |
    // ignore_for_file: camel_case_types, non_constant_identifier_names,
unused_element, unused_field, return_of_invalid_type, void_checks,
annotate_overrides, no_leading_underscores_for_local_identifiers,
library_private_types_in_public_api
```

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Direct platform library interop

ffigen

Objective-C

Swift

```
import 'dart:ffi';
import 'swift_api_bindings.dart';

void main() {
  final lib = SwiftLibrary(DynamicLibrary.open('libswiftapi.dylib'));
  final object = Example.new1(lib);
  print(object.sayHello(11,33));
  print('field = ${object.someField}');
  object.someField = 456;
  print('field = ${object.someField}');
}
```

New Architectures

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New Architectures

Null Safety

100% Sound Null Safety

100% Sound Null Safety

Dart 1.0

```dart
int getAge(Pearson p){
  return p.age;
}
```

```
;; Prologue
movq r13,[r12+0x17]
;; Invocation Count Check
movq rti,[r13+0x17
incl [rdi+0x77]
cmpl [rdi+0x77], 0x2326
jl 0x101f29e04
movq r12,[r13+0x1f]
movq r11,[r12+0x7]
jmp r11
;; Enter frane
push rbp
movq rbp,rsp
push r12
push pp
rovq pp,r13
subq rsp, 0x10
;; Initialize spill slots
movq rax,[thr+0x60] null
movq [rbp-0x18], rax
movq r11, [pp+0x27]
movq [rbp-0x1f],2
;; Edge counter
movq rax,[pp-0x2f]
addq [rax+0x1f],2
;; DebugStepCneck:4()
movq r12.[pp+0x37]
movq r11,[r12+0x7]
call r11
;; CheckStackOverflow: 6()
cmpq rsp,[thr+0x8]
jna exlelf29e60
;; t8 <- LoadLocal(a 32)
push [rbp+0x10]
;; PushArgument(t0)
movg rbx,[pp+0x3f]
movo r12,[pp+0x47]
moot r11,[r12+Ox7]
call r11
pop fish
push rax
;; DebugStepCeck:10()
movq r12,[pp+0x4f]
movo r11.[r12+0x7]
call r11
;; Return:12(t0)
pop rax
movq pp,[rbp-0x10]
movq rsp,rbp
pop rbp
retl
int3l
;; ChetkStackOverflow
movq rbx,[thr+0x4f]
movq r10,0
movq r11,[thr+0x10]
movq r12,[tnr+0x7]
call r11
jmp ax101f29e4b;
```

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100% Sound Null Safety

Dart 2.0

```
int getAge(Pearson p){
  return p.age;
}
```

```
;; Prologue
push rbp
movq rbp,rsp
movq rcx,[rbp+0x10]

;; Null check
cmpq rcx,[thr+0xd0] null
jz 0x0000000108e956ae

;; Field access
movq rax,[rcx+0x7]

;; Epilogue
movq rsp,rbp
Apri tutto
ret

;; Null handler
call 0x0000000108e956ae
```

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100% Sound Null Safety

Dart 3.0

```
;; Prologue
movq rcx,[rsp+0x8]

;; Field access
movq rax,[rcx+0x7]

;; Return
ret boris
```

```dart
int getAge(Pearson p){
  return p.age;
}
```

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Flutter

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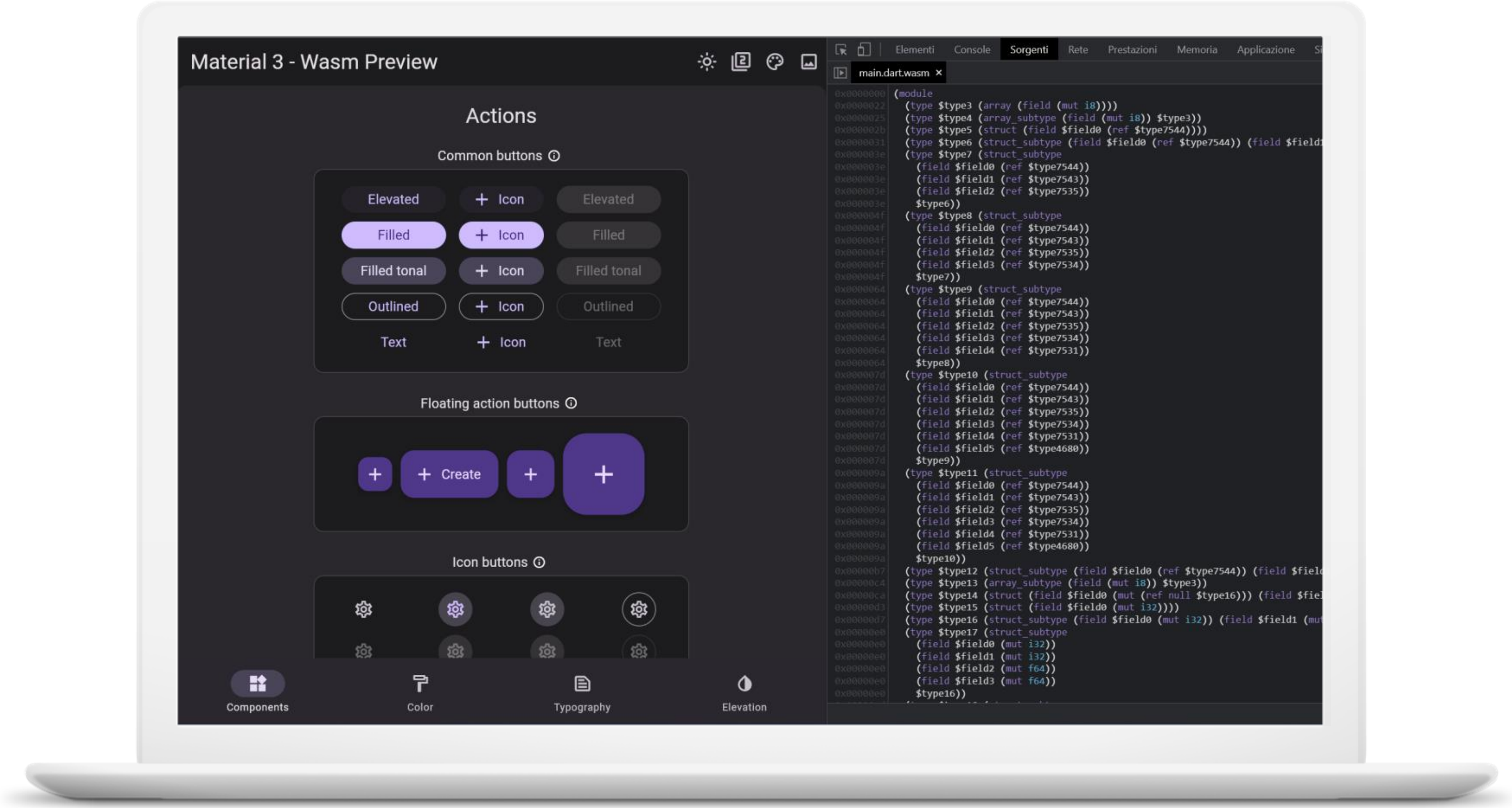
WebAssembly Support

WebAssembly Support



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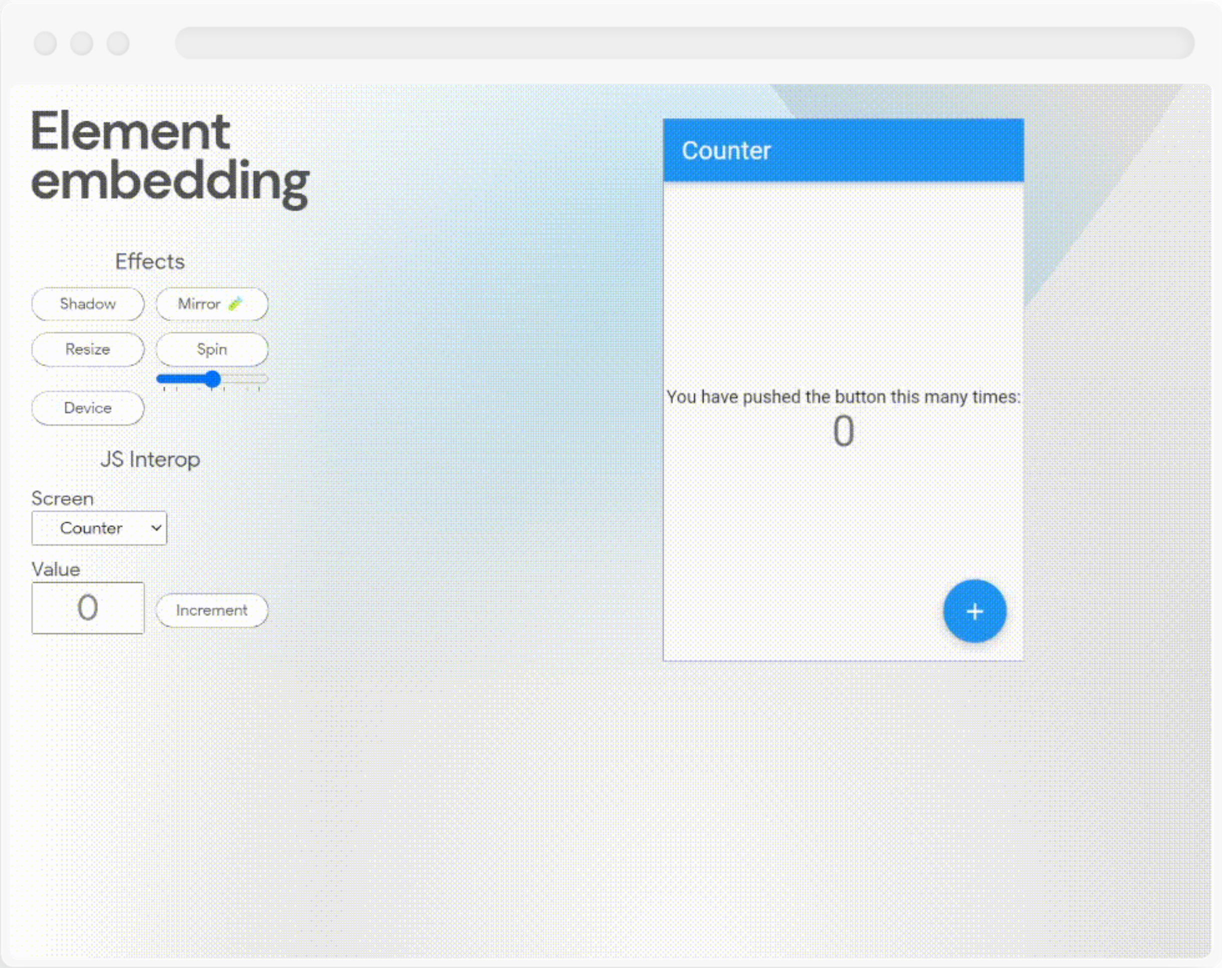
https://flutterweb-wasm.web.app/#/

Embedding

Embedding



https://flutter-forward-demos.web.app/#/

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Impeller

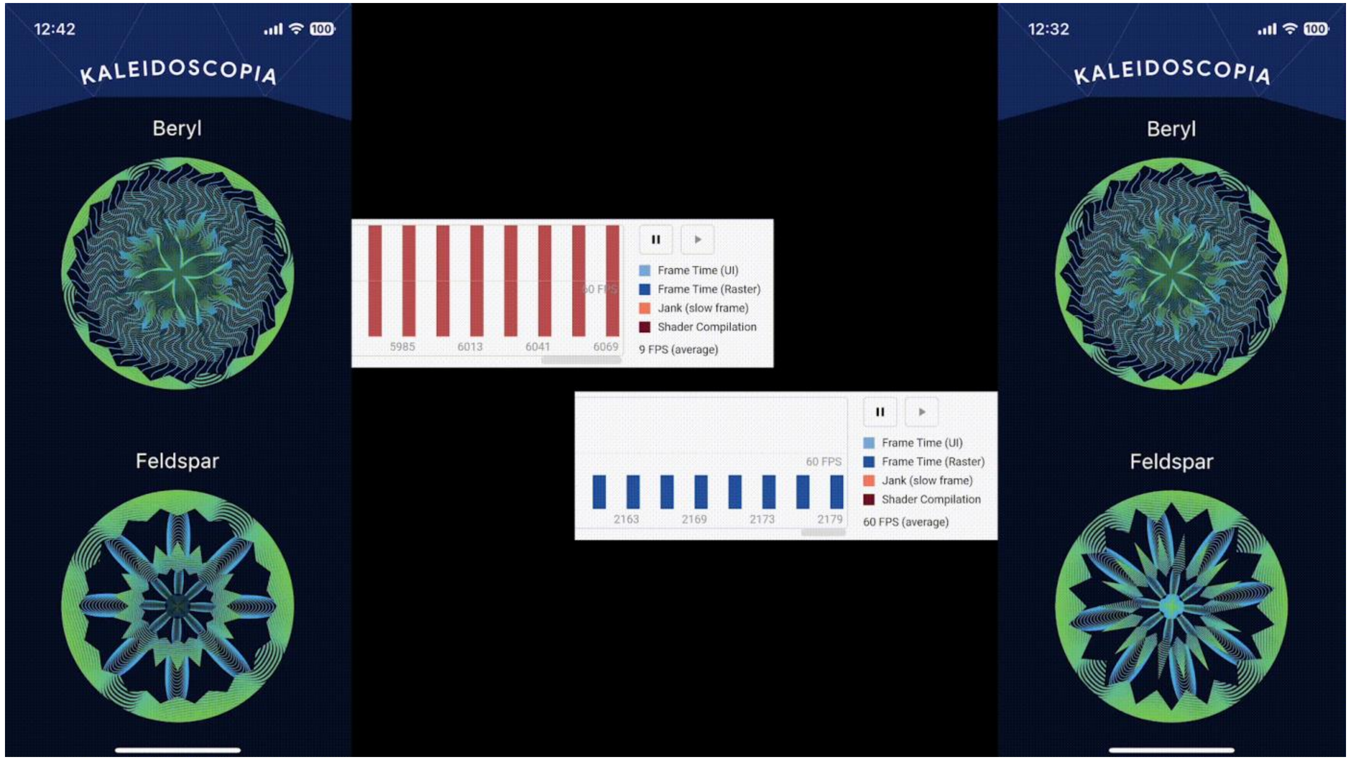
Impeller

Nuovo rendering runtime che andrà a sostituire Skia

- **Compilazione** degli shader e reflection realizzati **durante il build dell'app**
- **Ciascuna risorsa grafica viene etichettata.** Le animazioni vengono catturate e salvate su disco senza intaccare le performance
- **Portatile** (gli shader non dipendono dalla piattaforma target)
- **Adozione effettiva delle API Metal e Vulkan**
- **Multithread rendering**

Impeller

Boost prestazionale



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Impeller (3D)

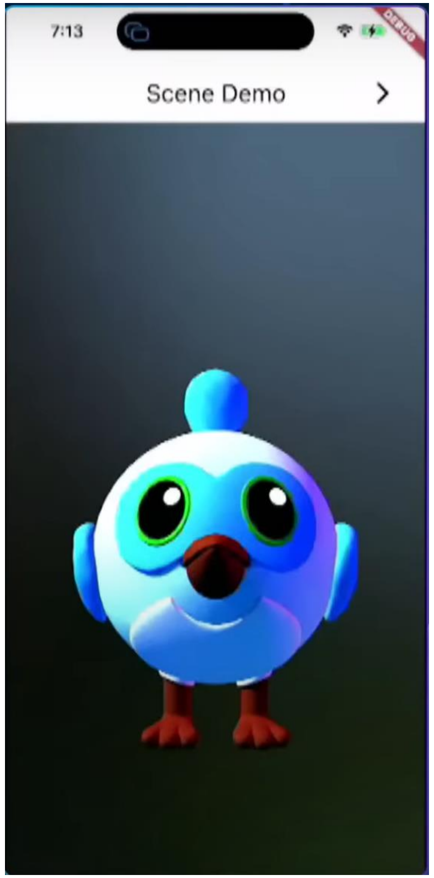
```dart
class DashWidget extends StatelessWidget {
  const DashWidget({super.key});

  @override
  Widget build(BuildContext context) {
    return Image(
      image: AssetImage(
        "assets/dash.png",
      ),
    );
  }
}
```

Impeller (3D)

```dart
class DashWidget extends StatelessWidget {
  const DashWidget({super.key});

  @override
  Widget build(BuildContext context) {
    return Scene(
      node: Node.asset(
        "models/dash.glb",
      ),
    );
  }
}
```



Impeller (3D)

```
class DashWidget extends StatelessWidget {
  const DashWidget({super.key});

  @override
  Widget build(BuildContext context) {
    return Scene(
      node: manyNodes(
        Node.asset(
          "models/dash.glb",
          animations: ["Walk"],
        ),
      ),
    );
  }
}
```

Domande?

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Grazie per l'attenzione

bonfry

bonfry.com

Lascia un feedback

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